# Networking, SASE & Palo Alto

Informatiks

Sitraka FORLER
sitrakaforler@gmail.com

*Centrale Méditerranée, AMU*

# How to Read These Slides

Every concept is presented across **two consecutive slides**:

**Kid Intuition**

**Slide 1 — Intuition**
A plain-English story with everyday analogies. **No jargon needed!**

**Engineer's View**

**Slide 2 — Engineer's View**
Protocols, RFCs, packet formats and vendor specifics. **Bring your terminal!**

CENTRALE MARSEILLE

## What Is a Network? — Intuition

### Kid Intuition

Imagine a city. Houses are **devices**. Roads are **cables** (or radio waves). The postal service decides *how* to get a letter from one house to another.

A **computer network** is exactly that city — devices connected by links, with agreed-upon rules (protocols) so they can exchange information reliably, even if some roads are blocked.

# What Is a Network? — Engineer's View

## Engineer's View

A network is a collection of **nodes** (hosts, routers, switches) connected by **links** (copper, fibre, wireless). Communication relies on **protocols**: agreed message formats and procedures.

Three fundamental metrics:

## Key Network Metrics

| Metric | Unit | Meaning |
|---|---|---|
| Bandwidth | bps | Max data rate of a link |
| Latency | ms | Time for one packet to traverse the path |

CENTRALE MARSEILLE

# The OSI Model — Intuition

## Kid Intuition

Sending a message over a network is like sending a parcel internationally. Different people handle different parts of the job:

**You** write the letter (Application). **A translator** puts it in a common language (Presentation). **The post office** opens an account for the delivery (Session). **The courier** splits it into parcels and tracks each one (Transport). **Logistics** finds the best route (Network). **The truck driver** takes it to the next hub (Data Link). **The road** physically carries the truck (Physical).

# The OSI Model — Engineer's View

## OSI Reference Model — ISO/IEC 7498

| Layer | Name | Protocol examples | PDU |
|---|---|---|---|
| 7 | Application | HTTP, DNS, SMTP | Data |
| 6 | Presentation | TLS, MIME | Data |
| 5 | Session | RPC, NetBIOS | Data |
| 4 | Transport | TCP, UDP | Segment |
| 3 | Network | IP, ICMP, OSPF | Packet |
| 2 | Data Link | Ethernet, Wi-Fi | Frame |
| 1 | Physical | Fibre, DSL, 5G | Bit |

## Encapsulation

CENTRALE MARSEILLE

# IP Addressing & Subnetting — Intuition

**Kid Intuition**

Think of an IP address like a postal address: **street name** (network part) + **house number** (host part). A **subnet mask** tells you how many bits belong to the street name.

CIDR notation `192.168.1.0/24` means: the first 24 bits identify the street, leaving 8 bits (256 houses) for devices on that street.

# IP Addressing & Subnetting — Engineer's View

## IPv4 CIDR

An IPv4 address is a 32-bit integer, written in dotted-decimal. A prefix `A.B.C.D/n` defines:

$$\text{Network address} = \text{IP } \texttt{AND} \text{ mask}, \quad \text{Hosts} = 2^{32-n} - 2.$$

## RFC 1918 — Private Ranges

| Range | Typical use |
|---|---|
| `10.0.0.0/8` | Large enterprise core |
| `172.16.0.0/12` | Medium enterprise / cloud VPCs |
| `192.168.0.0/16` | Branch / home networks |

CENTRALE MARSEILLE

# Routing & Switching — Intuition

## Kid Intuition

A **switch** is a smart power strip inside one building: it learns which socket each device is on, and only sends the message to the right socket (not to everyone at once).

A **router** is a GPS navigation system between cities. It knows the map of the internet and chooses the fastest or cheapest route to forward each packet toward its destination.

CENTRALE MARSEILLE

# Routing & Switching — Engineer's View

## Layer-2 Switching

Forwards **Ethernet frames** by **MAC address** using a CAM table.
Loop prevention: `STP`/`RSTP` (IEEE 802.1D/w).
Segmentation: `VLANs` (IEEE 802.1Q).

## Layer-3 Routing

Forwards **IP packets** using a **FIB** (Forwarding Info Base).
**Static routes** — manual.
**OSPF** — link-state IGP.
**BGP** — path-vector EGP (the internet's glue).

## Longest Prefix Match

When multiple routes match a destination, the router always

CENTRALE MARSEILLE

## Why Network Security? — Intuition

**Kid Intuition**

Imagine leaving your front door wide open in a city. Anyone can walk in, take your belongings, or pretend to be you. Network security is the lock, the alarm system, the security camera, and the bouncer — all working together.

The internet is that city, and without security, every device on your network is that open door.

CENTRALE MARSEILLE

# Why Network Security? — Engineer's View

## The CIA Triad

| Property | Threat if violated |
|---|---|
| **C**onfidentiality | Eavesdropping, data exfiltration |
| **I**ntegrity | Tampering, man-in-the-middle |
| **A**vailability | DoS/DDoS, ransomware |

## Defence in Depth

No single control is sufficient. Secure architectures layer multiple controls: **perimeter firewall** → **IDS/IPS** → **endpoint protection** → **SIEM/SOAR**.

CENTRALE MARSEILLE

## Firewalls — Intuition

### Kid Intuition

A **firewall** is a security guard at the gate of your building. It checks every person (packet) trying to enter or leave.

An old-school guard only checks IDs (IP addresses and port numbers). A **Next-Generation Firewall (NGFW)** — like Palo Alto's — is a trained detective: it actually reads what's inside the bag (the payload), identifies the application, and checks the person's behaviour history before deciding to let them through.

CENTRALE MARSEILLE

# Firewalls — Engineer's View

## Firewall Generations

| Generation | Inspection | Limitation |
|---|---|---|
| Packet filter | IP/port (L3-L4) | No state, evasion-prone |
| Stateful | TCP session table | No app visibility |
| Proxy / ALG | Application protocol | Performance cost |
| NGFW | App-ID + User-ID + Content | Requires skilled tuning |

## Palo Alto NGFW — Single-Pass Architecture

PAN-OS processes each packet **once** through a parallel pipeline: App-ID ‖ User-ID ‖ Content-ID ‖ Decryption. This avoids the performance penalty of stacking multiple engines.

CENTRALE MARSEILLE

## Zero Trust — Intuition

### Kid Intuition

Old security said: *"If you're inside the office building, you're trusted."* The problem? Once an attacker gets inside (via phishing, a stolen laptop, a rogue USB), they can roam freely.

**Zero Trust** says: *"Never trust, always verify."* Even if you're already inside, you must prove who you are and what you need — **every single time**, for every resource.

# Zero Trust — Engineer's View

## NIST SP 800-207 — Zero Trust Principles

1. All data sources and services are considered **resources**.
2. **All communication is secured** regardless of network location.
3. Access is granted **per-session**, on a least-privilege basis.
4. Access is determined by **dynamic policy**: identity + device posture + context.
5. The enterprise **monitors and measures** the integrity of all assets.

## Palo Alto Zero Trust Enablers

**User-ID** (identity), **GlobalProtect** (device posture), **Cortex XDR** (behaviour), **Prisma Access** (cloud enforcement

CENTRALE MARSEILLE

## What is SASE? — Intuition

### Kid Intuition

Historically, all roads led to the corporate office: employees connected to the office network, which led to the data centre, which held all the apps.

Now apps live in the cloud, employees work from home, and those roads no longer make sense. **SASE** (pronounced "sassy") says: instead of routing everything back to the office, bring the **security checkpoint directly to the user**, wherever they are — like a passport control booth that pops up next to you, no matter which country you're in.

CENTRALE MARSEILLE

## What is SASE? — Engineer's View

**SASE — Gartner, 2019**

**Secure Access Service Edge** converges **WAN networking** and **network security** into a *single cloud-delivered service*, consumed as-a-service and enforced as close to the user/device as possible.

### Network Edge (WAN)

- SD-WAN
- WAN Optimisation
- QoS / Traffic shaping

### Security Edge (SSE)

- ZTNA
- SWG (Secure Web Gateway)
- CASB
- FWaaS

## SASE Components Deep Dive — Intuition

### Kid Intuition

Think of SASE as a Swiss Army knife for network security:

**ZTNA** = The "prove who you are before opening any door" tool.
**SWG** = A filter that blocks dangerous websites, like parental controls for companies.
**CASB** = A watchdog that monitors what employees do in cloud apps (Salesforce, OneDrive...).
**FWaaS** = Your NGFW, but running in the cloud instead of a physical box in your office.

# SASE Components Deep Dive — Engineer's View

## SSE Components

| Component | Function | Key standard/tech |
|---|---|---|
| ZTNA | App-level remote access | BeyondCorp, mTLS |
| SWG | Outbound web filtering | URL cat., SSL inspection |
| CASB | SaaS visibility/control | API mode, inline proxy |
| FWaaS | Cloud-delivered L7 FW | NGFW in PoP |
| DEM | Digital experience monitoring | Synthetic probes |

## SSE vs. SASE

**SSE** (Security Service Edge) = security-only subset of SASE, without SD-WAN. Gartner split the market in 2021. Palo

## SD-WAN — Intuition

**Kid Intuition**

A traditional WAN is like renting a dedicated private road between your offices — very reliable, very expensive. SD-WAN is like using **Google Maps on a public highway**: it watches all available roads (fibre, 4G, MPLS), picks the best one in real-time, and automatically reroutes if there is a traffic jam.

You get near-private-road quality at public-road prices.

# SD-WAN — Engineer's View

## SD-WAN Key Capabilities

- **Transport independence:** bond MPLS + broadband + LTE into one logical link.
- **Application-aware routing:** steer traffic per-app based on SLA thresholds (latency, jitter, loss).
- **Zero-touch provisioning (ZTP):** branches self-configure from the cloud; no on-site engineer required.
- **Centralised orchestration:** single control plane manages all sites.

## Palo Alto Prisma SD-WAN (formerly CloudGenix)

Prisma SD-WAN integrates natively with Prisma Access: traffic from branches is automatically steered to the near-

CENTRALE MARSEILLE

# PAN-OS Architecture — Intuition

**Kid Intuition**

Most old firewalls are like a slow assembly line: each worker checks one thing and passes the box on. The box goes through 10 workers before it leaves.

**PAN-OS** is like a team of specialists who all inspect the same box *simultaneously*, in parallel. The result is out the door far faster, and nothing is missed.

That is Palo Alto's famous **Single-Pass Parallel Processing (SP3)** engine.

# PAN-OS Architecture — Engineer's View

## SP3 — Single-Pass Parallel Processing

1. **App-ID**: identifies the application using signatures, decoders, and heuristics (not port/protocol alone).
2. **User-ID**: maps IP address to username via AD, LDAP, Captive Portal, or API.
3. **Content-ID**: inspects payload for threats (AV, IPS, URL filtering, DLP, WildFire sandbox).
4. **Decryption**: TLS/SSH inspection inline, before policy evaluation.

## Policy Model

Rules are evaluated in order: `Zone` → `Source/Dest IP/User` → `App-ID` → `Service` → `Action`

CENTRALE MARSEILLE

## WildFire — Intuition

**Kid Intuition**

What happens when a threat is brand new and no one has seen it before? Traditional antiviruses only recognise known criminals. **WildFire** is the forensics lab: it takes suspicious files, locks them in a safe room (a sandbox), and watches what they actually *do*. If they behave like malware, the verdict is shared with every Palo Alto firewall on the planet within **5 minutes**.

CENTRALE MARSEILLE

# WildFire — Engineer's View

## WildFire Analysis Pipeline

1. Firewall forwards unknown file (PE, PDF, Office, APK. . . ) to WildFire cloud.
2. File is detonated in **multi-OS VMs** (Windows, macOS, Android, Linux).
3. **Static analysis** (signatures, ML) + **dynamic analysis** (API calls, network, registry).
4. Verdict: `benign` | `grayware` | `malware` | `phishing`.
5. **Signature pushed globally in** $\leq 5$ **min** (real-time WildFire subscribers).

## Inline ML — PAN-OS 10.0+

For zero-day prevention *without* cloud round-trip, Palo Alto

CENTRALE MARSEILLE

## Prisma Access — Intuition

### Kid Intuition

Imagine your office's security guard. Great for people who come to the office. But what about remote workers, travelling employees, and cloud apps?

**Prisma Access** creates **virtual security guard posts** in over 100 cities around the world. No matter where your employee is, their traffic passes through the nearest guard post — which has all the same NGFW rules as your physical office firewall.

# Prisma Access — Engineer's View

## Prisma Access Architecture

- **PoPs (Points of Presence):** 100+ locations, Palo Alto-managed, running PAN-OS in cloud.
- **Remote users:** connect via GlobalProtect agent (IPsec/TLS) to nearest PoP.
- **Branch/HQ:** connect via IPsec tunnels (service connection) or Prisma SD-WAN.
- **Enforcement:** same Security, QoS, and Decryption policies as on-prem NGFW.
- **Management:** Panorama / Strata Cloud Manager (SCM) — single pane of glass.

## Autonomous Digital Experience Management (ADEM)

CENTRALE MARSEILLE

## Cortex XDR — Intuition

### Kid Intuition

Traditional security products are like separate security cameras in different buildings — each watches its own corner, but nobody is looking at all the cameras at once.

**Cortex XDR** is the **unified security operations centre**: it receives feeds from all cameras (firewall, endpoint, cloud, identity), stitches them into a single timeline of events, and automatically spots when an attacker is quietly moving from room to room.

CENTRALE MARSEILLE

# Cortex XDR — Engineer's View

## XDR — Extended Detection & Response

- **Data sources:** Firewall logs, WildFire, GlobalProtect, Cortex agent (EDR), SaaS APIs, 3rd-party SIEM ingestion.
- **Stitching:** causality chain analysis correlates `Alert` → `Incident` across sources.
- **BIOC (Behavioral Indicators of Compromise):** rule-based + ML detections on raw telemetry.
- **Response:** isolate endpoint, block IP/hash on NGFW, quarantine file — all from one console.

## MITRE ATT&CK Mapping

Cortex XDR maps every alert to the **MITRE ATT&CK** framework tactic/technique, enabling SOC teams to immedi-

CENTRALE MARSEILLE

# Traditional vs. SASE Architecture — Intuition

**Kid Intuition**

**Traditional (hub & spoke):**
All branch traffic drives into HQ data centre, gets inspected, then goes to the internet. Fast for internal apps, slow and expensive for cloud.

**SASE (cloud-delivered):**
Each user/branch connects directly to the nearest SASE PoP. The PoP applies all security controls, then forwards traffic optimally — straight to the cloud app, not via HQ. Faster, cheaper, more scalable.

# Traditional vs. SASE Architecture — Engineer's View

## Traditional WAN

Branch $\xrightarrow{\text{MPLS}}$ HQ DC $\xrightarrow{\text{NGFW}}$ Internet

**Pros:** centralised control.
**Cons:** hairpinning, MPLS cost, latency to SaaS, VPN complexity.

## SASE / Prisma Access

Branch / User $\xrightarrow{\text{IPsec/GP}}$ Prisma PoP $\rightarrow$ Internet / SaaS

**Pros:** local breakout, elastic scale, unified policy.
**Cons:** vendor trust, internet dependency.

## Palo Alto SASE Stack

**Prisma SD-WAN** + **Prisma Access** + **GlobalProtect** + **Cortex XDR** = Full SASE with native NGFW-grade enforcement and unified visibility

# Deployment Checklist — Engineer's View

## Engineer's View

### Phase 1 — Visibility

- Enable App-ID and User-ID in `monitor` mode (no blocking yet).
- Activate **SSL decryption** for key categories.
- Onboard logs to **Cortex Data Lake**.

### Phase 2 — Enforcement

- Create **Security Zones**: trust / untrust / DMZ / cloud.
- Migrate from port-based rules to **App-ID rules**.
- Attach Security Profiles: AV, Vulnerability, WildFire, URL, DNS.

### Phase 3 — SASE Rollout

- Deploy **Prisma Access** PoP connections; migrate VPN

## Summary — Intuition

### Big Picture

1. **Networking:** devices talking via agreed protocols over layered stacks (OSI).
2. **Security:** firewalls evolved from port-filters to NGFW app-aware engines.
3. **Zero Trust:** never trust location; always verify identity and posture.
4. **SASE:** merge WAN + security into a cloud service next to the user.
5. **Palo Alto NGFW:** single-pass, App-ID + User-ID + Content-ID.
6. **Prisma Access:** NGFW-in-the-cloud for every user, everywhere.

# Summary — Engineer's View

## Technical Core

| Concept | Key takeaway |
|---|---|
| OSI model | 7-layer encapsulation; know L2/L3/L4/L7 |
| CIDR subnetting | $2^{32-n} - 2$ hosts; longest prefix wins |
| NGFW / SP3 | App-ID + User-ID + Content-ID in one pass |
| Zero Trust | NIST SP 800-207; per-session, least-privilege |
| SASE | SD-WAN + SSE (ZTNA + SWG + CASB + FWaaS) |
| Prisma Access | 100+ PoPs; GlobalProtect; no hairpin |
| WildFire | Sandbox + ML; verdict in $\leq 5$ min globally |
| Cortex XDR | XDR + BIOC + MITRE ATT&CK mapping |

## Key References


- Tanenbaum, A. & Wetherall, D. (2011). *Computer Networks*, 5th ed. Pearson.
- Gartner (2019). *The Future of Network Security Is in the Cloud* (SASE coinage).
- NIST SP 800-207 (2020). *Zero Trust Architecture*.
- RFC 4271 — *A Border Gateway Protocol 4 (BGP-4)*.
- RFC 2328 — *OSPF Version 2*.
- Palo Alto Networks (2024). *PAN-OS Administrator's Guide, v11.x*.
- Palo Alto Networks (2024). *Prisma Access Administrator's Guide*.
- MITRE (2023). *ATT&CK Enterprise Matrix v14*.